

Programming 1: Lab 2 : Operators, Expressions and Conditions

Write the python code for the following questions.

1. Write a program to display “Lucky” if the number entered by user is divisible by 7, otherwise print “Unlucky”.
2. If the length and breadth of a room are L meters and B meters, respectively, and the height is H meters, what is the area of its four walls? The area of the four walls is given by $2 \times H \times (L + B)$. Write Python code to output the area by taking L, B and H as input from the user.
3. Write a program that interprets the rainfall condition of a region based on the total rainfall received and the number of days over which it fell.

It should tell them the interpretation based on the average daily rainfall value.

- Under 2 mm it is a Drought
- Over 2 mm but below 5 mm the rainfall is Deficient
- Over 5 mm but below 10 mm the rainfall is Normal
- Over 10 mm but below 20 mm the rainfall is Excess
- Above 20 mm it is a Flood alert

The average daily rainfall is calculated by dividing the total rainfall (in mm) by the number of days. Take the total rainfall and the number of days from the user.

4. Write a Python program to check whether a character entered by the user is a digit, an alphabet or a special symbol.
5. Suppose a town initially had a population of 3.45×10^5 people. After some time, the town had 4.12×10^5 people. Calculate the percentage of population growth. Display the answer rounded to the nearest whole number.
6. Write a program that asks the user to enter a whole number of paise as input and then converts that amount of money to rupees and paise. The program should use both integer division and the modulus operator.
7. The fare of a metro ride is decided by the type of card and the time of travel according to the following rule:
Student card, peak hours, twenty rupees.
Student card, off-peak hours, fifteen rupees.
Regular card, peak hours, thirty-five rupees.
Regular card, off-peak hours, twenty-five rupees.
Write a program that requests a card type (Student or Regular) and a travel time (Peak or Offpeak) as input and then displays the fare.
8. Write a program that requests the scores of two teams in a quiz and displays the message “Match drawn” if the scores are equal; otherwise, it displays which team won and the margin of victory.
9. A photocopy shop charges 200 paise per page for small jobs or 150 paise per page for jobs of 50 pages or more. Write a program that requests the number of pages ordered and displays the total cost.

10. A petrol pump sells petrol for Rs. 96.50 per litre. Write an attendant's program that requests the number of litres filled and the amount of cash tendered as input and displays the change from the transaction. If the cash is not enough, the message "You owe Rs. x.xx more." should be displayed, where Rs. x.xx is the difference between the total cost and the cash.
11. Find the smallest of 3 numbers using nested if else condition.
12. Calculate the total hotel bill of a guest by taking as input the room rent. Service charge = 10% of room rent, GST is 12% of room rent, and tourism cess is 3% of room rent. Total Bill = Room Rent + Service Charge + GST + Tourism Cess.
13. Further, in the above question, find the loyalty discount slab in which the guest lies.

Total Bill

< Rs. 5,000

Rs. 5,000 - Rs. 15,000

Rs. 15,000 - Rs. 40,000

Above Rs. 40,000

Loyalty Discount

0%

5% of the total bill

8% of the total bill

12% of the total bill

14. The flowchart given below calculates a courier company's delivery charge on the basis of the weight of the parcel in kilograms. Write a program corresponding to the flowchart.

